HW 1 — Random Variables, Joints, Conditionals, Bayes

Mathematical Foundations of Diffusion Models · Monsoon 2026 · IIT Jammu

Released	Monday 10 August 2026
Due	Tuesday 18 August 2026, in class
Marks	50 points — graded for completion

Graded for completion. Credit is for a genuine attempt at every part, not for arriving at the right answer. Attempt everything, and write down your reasoning even where you are unsure — a wrong answer with visible working earns full credit, a blank does not. Solutions are posted after the deadline; check your own work against them, and bring anything that still does not make sense to class.

Show your working. Give exact fractions and decimals to three places.

Problem 1

Roll one fair six-sided die, once. Before the questions, here is what each symbol in them means.

- Ω is the **sample space**: the set of every outcome the experiment could produce. For this die an outcome is just the face that comes up.
- ω is a single **outcome** — one particular thing that happened, one element of Ω. Exactly one ω occurs each time you roll.
- An **event** is any subset $A \subseteq \Omega$: a yes/no question about the outcome. "The face is even" is the event $\{2, 4, 6\}$, and it occurs precisely when the ω that happened lies inside it.
- A **random variable** is a function $X : \Omega \to \mathbb{R}$ that attaches a number to *each outcome*. It is not random and it is not a variable: it is an ordinary deterministic function. All the randomness sits in which ω turns up.

Two random variables on this die:

- $X(\omega) = \omega$ — read the face and report it as a number. So $X(4) = 4$. (This one is the identity function; it looks like it is doing nothing, and that is the point: the outcome already *is* a number here, which is not true in general.)
- $Y(\omega) = 1$ if ω is prime, and 0 otherwise — report whether the face is prime. So $Y(3) = 1$ and $Y(4) = 0$.

(a) Write down Ω explicitly, and say how many distinct events this experiment has. Justify the count in one line. **(2pts)**

(b) List the two sets $\{\omega : Y(\omega) = 1\}$ and $\{\omega : Y(\omega) = 0\}$. (Read the notation as "all outcomes ω for which Y returns that value".) **(2pts)**

(c) A classmate says "Y maps the event $\{2, 3, 5\}$ to 1". Explain why this is the wrong description of what Y does, given what a random variable takes as its input. **(3pts)**

(d) Compute $\mathbb{P}(Y = 1)$ and $\mathbb{P}(X \le 3,\ Y = 1)$. In each case name the event whose probability you are computing before you compute it. **(3pts)**

Total: 10 points

Problem 2

A workshop runs three machines, A, B and C, which all make the same part. Every finished part is inspected and given one of three quality grades:

Grade	Meaning
1	within tight tolerance — ships as premium
2	within tolerance — ships as standard
3	out of tolerance — sent for rework

A day's output of 200 parts is inspected, and each part is recorded twice over: which machine made it, and what grade it received. The counts are

	grade 1	grade 2	grade 3
machine A	40	20	10
machine B	15	35	20
machine C	10	15	35

Now pick one of the 200 parts uniformly at random, and let $X \in \{A, B, C\}$ be the machine that made it and $Y \in \{1, 2, 3\}$ its grade. Both are random variables on the same outcome — the part you picked.

The question behind the arithmetic is whether the machine a part came from tells you anything about the grade it is likely to get. If it does, the workshop has a machine worth investigating.

(a) Compute the row totals, column totals, and both marginal distributions $\mathbb{P}(X = \cdot)$ and $\mathbb{P}(Y = \cdot)$. **(2pts)**

(b) Compute $\mathbb{P}(Y = \cdot \mid X = A)$ and $\mathbb{P}(Y = \cdot \mid X = C)$. Comparing the two, does knowing the machine change what you expect of the grade — that is, are X and Y dependent? **(3pts)**

(c) A part is pulled off the standard-grade pile, so you know only that $Y = 2$. Compute $\mathbb{P}(X = B \mid Y = 2)$, stating which total you divided by and why it is not 200. **(2pts)**

(d) If machine and grade were unrelated, how many of the 200 parts would you expect to be grade-1 parts from machine A? Compare with the observed count, and say what the comparison suggests. **(3pts)**

Total: 10 points

Problem 3

A university holds a campus health camp and offers every student a quick finger-prick blood-glucose screen for **type 2 diabetes**. Among students of this age group roughly **1 in 100** has the condition, very often without knowing it.

The screen is not a diagnosis; it only flags who should be sent for a proper fasting-glucose or HbA1c test. Suppose it behaves as follows:

- it is positive for 95% of students who *do* have type 2 diabetes;
- it is also positive for 10% of students who do *not* — a single high reading can follow a recent meal, an illness, poor sleep, or ordinary day-to-day variation.

Let D be the event "the student has type 2 diabetes" and $+$ the event "the screen reads positive". A student is picked at random from those screened.

(a) Write the three given numbers in conditional-probability notation. **(2pts)**

(b) Compute $\mathbb{P}(+)$: the fraction of screened students who get a positive reading. **(3pts)**

(c) Compute $\mathbb{P}(D \mid +)$: given a positive reading, the probability the student actually has the condition. **(3pts)**

(d) The screen catches 95% of true cases, yet your answer to (c) is far below 95%. Explain why. Then say which of the three given numbers you would change, and in which direction, to raise $\mathbb{P}(D \mid +)$ the most. **(2pts)**

Check: redo (b) and (c) by imagining the camp screens 10,000 students, and filling in a 2×2 *table of counts. Same answer, easier arithmetic.*

The figures above are illustrative and chosen to keep the arithmetic clean; they are not the published operating characteristics of any particular test. The structure, though, is real, and it is exactly why a positive screening result triggers a confirmatory test rather than a diagnosis.

Total: 10 points

Problem 4

Let (X, Y) have joint density $f(x, y) = c\,(x + y)$ on $0 \le x \le 1$, $0 \le y \le 2$, and $f = 0$ elsewhere.

(a) Find c. **(2pts)**

(b) Find $f_X(x)$ and $f_Y(y)$ with their ranges, and verify each integrates to 1. **(3pts)**

(c) Find $f(y \mid x)$ and verify it integrates to 1 over y. **(3pts)**

(d) Are X and Y independent? Justify by explicit comparison. **(2pts)**

Total: 10 points

Problem 5

Two machines feed the same conveyor. Machine 1 is the older unit and runs 60% of the output; machine 2 runs the rest. At the end of the line every item is inspected and its surface defects counted: 0, 1 or 2.

Here is the difference from Problem 2, and it is the whole point. There, the machine was written on the inspection sheet, so you observed *both* variables and could build the full table. Here the two machines' output mixes on the shared conveyor, and by the time an item reaches inspection nobody knows which machine made it. So the machine label Z is **latent** — never observed — while the defect count X is all you get to see.

It still matters. If one machine is drifting out of adjustment you want to know which one, so you can stop the line and service it; but the only evidence available is defect counts. Recovering Z from X is an inference problem, and Bayes' rule is what does it.

The two machines have different defect profiles:

	$X = 0$	$X = 1$	$X = 2$
$\mathbb{P}(X = \cdot \mid Z = 1)$	0.7	0.2	0.1
$\mathbb{P}(X = \cdot \mid Z = 2)$	0.2	0.3	0.5

Pick an inspected item at random. Write $Z \in \{1, 2\}$ for the machine that made it and $X \in \{0, 1, 2\}$ for its defect count. You see X; you want Z.

(a) Using $p(z, x) = p(z)\, p(x \mid z)$, write the full 2×3 joint table and check it sums to 1. **(2pts)**

(b) Compute $\mathbb{P}(X = x)$ for each x, stating which variable you summed out. This is the defect distribution the inspector actually sees, with the machine identity averaged away. **(2pts)**

(c) An item is found with x defects. Compute $\mathbb{P}(Z = 1 \mid X = x)$ for each of $x = 0, 1, 2$ — the belief that the older machine made it. **(3pts)**

(d) Compute $\mathbb{E}[Z \mid X = x]$ for each x, and explain why these values need not lie in $\{1, 2\}$ — and so what such a number is telling the engineer. **(2pts)**

(e) Verify that $\mathbb{E}\big[\mathbb{E}[Z \mid X]\big] = \mathbb{E}[Z]$. **(1pt)**

Parts (c) and (d) are, in miniature, what a diffusion model's denoiser computes: a posterior over a hidden quantity given a noisy observation, then its mean. The continuous version $\mathbb{E}[x_0 \mid x_\sigma]$ appears in the tutorial "The World's Smallest Diffusion Model".

Total: 10 points